Now, $\dfrac{z_1}{z_2}=\dfrac{(4,2)}{(3,-1)}=\dfrac{4+2i}{3-i}$

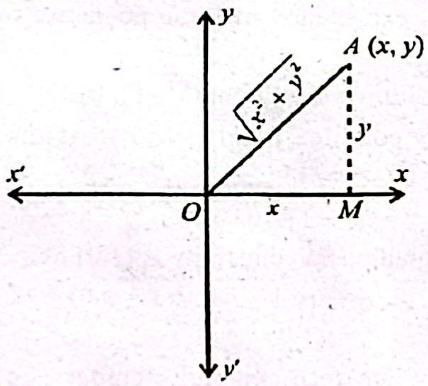
Multiply the numerator and denominator by the complex conjugate of $z_2 = 3 - i$.

$$\frac{z_1}{z_2}=\frac{4+2i}{3-i}=\frac{4+2i}{3-i}\times\frac{3+i}{3+i}$$
$$=\frac{(4)(3)+(4)(i)+(2i)(3)+(2i)(i)}{(3)^2-(i)^2}$$
$$=\frac{12+4i+6i+2i^2}{9-i^2}$$
$$=\frac{12+10i-2}{9-(-1)}=\frac{10+10i}{10}$$
$$=1+i \qquad \because\ i^2=-1$$

Thus, $\dfrac{z_1}{z_2}=1+i$

Modulus of Complex Number: The real number $\sqrt{x^2+y^2}$ is called the modulus of the complex number $x+iy$ and it is denoted by $|x+iy|$. In Figure, $|\overline{OA}|$ represent the modulus of $x+iy$. OR

In other words, the modulus of a complex number is the distance from the origin to the point representing the number.



Example 3: If $z=\dfrac{(1+2i)^2}{2-i}$ then evaluate $|\bar{z}|$

Sol. $z=\dfrac{(1+2i)^2}{2-i}=\dfrac{1+4i+4i^2}{2-i}$
$$=\frac{-3+4i}{2-i}\times\frac{2+i}{2+i}=\frac{-6-3i+8i+4i^2}{2^2-i^2}$$
$$=\frac{-6+5i-4}{4-(-1)}=\frac{-10+5i}{5}$$
$$\Rightarrow\quad z=-2+i$$

Taking conjugate $\bar{z}=\overline{-2+i}=-2-i$

and $|\bar{z}|=|-2-i|=\sqrt{(-2)^2+(-1)^2}=\sqrt{4+1}$

$\Rightarrow\ |\bar{z}|=\sqrt{5}$

Note:

- Every real number is a complex number with 0 as its imaginary part.
- A real number is self-conjugate.
- The set C of complex numbers does not satisfy the order axioms. In fact, there is no sense in saying that one complex number is greater or less than the other.

Exercise 1.1

Q1. Find the multiplicative inverse of each of the following numbers:

(i) $(-4, 7)$

Sol. Given that $z = (-4, 7)$

$\Rightarrow\ z=-4+7i$

So, $\dfrac{1}{z}=\dfrac{1}{-4+7i}\Rightarrow\dfrac{1}{z}=\dfrac{1}{-4+7i}\times\dfrac{-4-7i}{-4-7i}$

$\dfrac{1}{z}=\dfrac{-4-7i}{(-4)^2-(7i)^2}\Rightarrow\dfrac{1}{z}=\dfrac{-4-7i}{16-49i^2}$

$\dfrac{1}{z}=\dfrac{-4-7i}{16-49(-1)}\Rightarrow\dfrac{1}{z}=\dfrac{-4-7i}{16+49}$

$\dfrac{1}{z}=\dfrac{-4-7i}{65}\Rightarrow\dfrac{1}{z}=-\dfrac{4}{65}-\dfrac{7}{65}i$

$\dfrac{1}{z}=\left(-\dfrac{4}{65},-\dfrac{7}{65}\right)$

(ii) $(\sqrt{2},-\sqrt{5})$

Sol. Given that $z=\left(\sqrt{2},-\sqrt{5}\right)$

$\Rightarrow\ z=\sqrt{2}-\sqrt{5}i$

So, $\dfrac{1}{z}=\dfrac{1}{\sqrt{2}-\sqrt{5}i}\Rightarrow\dfrac{1}{z}=\dfrac{1}{\sqrt{2}-\sqrt{5}i}\times\dfrac{\sqrt{2}+\sqrt{5}i}{\sqrt{2}+\sqrt{5}i}$

$\dfrac{1}{z}=\dfrac{\sqrt{2}+\sqrt{5}i}{\left(\sqrt{2}\right)^2-\left(\sqrt{5}i\right)^2}\Rightarrow\dfrac{1}{z}=\dfrac{\sqrt{2}+\sqrt{5}i}{2-5i^2}$

$\dfrac{1}{z}=\dfrac{\sqrt{2}+\sqrt{5}i}{2-5(-1)}\Rightarrow\dfrac{1}{z}=\dfrac{\sqrt{2}+\sqrt{5}i}{2+5}$

$\dfrac{1}{z}=\dfrac{\sqrt{2}+\sqrt{5}i}{7}\Rightarrow\dfrac{1}{z}=\dfrac{\sqrt{2}}{7}+\dfrac{\sqrt{5}}{7}i\Rightarrow\dfrac{1}{z}=\left(\dfrac{\sqrt{2}}{7},\dfrac{\sqrt{5}}{7}\right)$

(iii) (1,0)

Sol. Given that $z = (1,0)$

$\Rightarrow\ z = 1 + 0i$

$z = 1$

So, $\dfrac{1}{z}=1\Rightarrow\dfrac{1}{z}=(1,0)$

Q2. **Separate into real and imaginary parts (write as a simple complex number):**

(i) $\dfrac{2-7i}{4+5i}$

Sol. Let $z=\dfrac{2-7i}{4+5i}$

$$=\frac{2-7i}{4+5i}\times\frac{4-5i}{4-5i}=\frac{2(4-5i)-7i(4-5i)}{(4)^2-(5i)^2}$$

$$=\frac{8-10i-28i+35i^2}{16-25i^2}$$

$$=\frac{8-10i-28i+35(-1)}{16-25(-1)}=\frac{8-10i-28i-35}{16+25}$$

$$=\frac{-27-38i}{41}=-\frac{27}{41}-\frac{38}{41}i$$

So, $\operatorname{Re}(z)=-\dfrac{27}{41}$ and $\operatorname{Im}(z)=-\dfrac{38}{41}$

(ii) $\dfrac{(-2+3i)^2}{1+i}$

Sol. let $z=\dfrac{(-2+3i)^2}{1+i}$

$$=\frac{(-2)^2+(3i)^2+2(-2)(3i)}{1+i}$$

$$=\frac{4+9i^2-12i}{1+i}=\frac{4+9(-1)-12i}{1+i}=\frac{4-9-12i}{1+i}$$

$$=\frac{-5-12i}{1+i}=\frac{-5-12i}{1+i}\times\frac{1-i}{1-i}$$

$$=\frac{-5(1-i)-12i(1-i)}{1^2-i^2}=\frac{-5+5i-12i+12i^2}{1-(-1)}$$

$$=\frac{-5+5i-12i+12(-1)}{1+1}$$

$$=\frac{-5+5i-12i-12}{2}=\frac{-17-7i}{2}=-\frac{17}{2}-\frac{7}{2}i$$

So, $\operatorname{Re}(z)=-\dfrac{17}{2}$ and $\operatorname{Im}(z)=-\dfrac{7}{2}$

(iii) $\dfrac{i}{1+i}$

Sol. Let $z=\dfrac{i}{1+i}$

$$=\frac{i}{1+i}\times\frac{1-i}{1-i}=\frac{i-i^2}{1^2-i^2}=\frac{i-(-1)}{1-(-1)}=\frac{i+1}{1+1}$$

$$=\frac{1+i}{2}=\frac{1}{2}+\frac{1}{2}i$$

So, $\operatorname{Re}(z)=\dfrac{1}{2}$ and $\operatorname{Im}(z)=\dfrac{1}{2}$

(iv) $\dfrac{(4+3i)^2}{4-3i}$

Sol. Let $z=\dfrac{(4+3i)^2}{4-3i}$

$$=\frac{(4)^2+(3i)^2+2(4)(3i)}{4-3i}=\frac{16+9i^2+24i}{4-3i}$$

$$=\frac{16+9(-1)+24i}{4-3i}=\frac{16-9+24i}{4-3i}=\frac{7+24i}{4-3i}$$

$$=\frac{7+24i}{4-3i}\times\frac{4+3i}{4+3i}=\frac{7(4+3i)+24i(4+3i)}{(4)^2-(3i)^2}$$

$$=\frac{28+21i+96i+72i^2}{16-9i^2}$$

$$=\frac{28+21i+96i+72(-1)}{16-9(-1)}=\frac{28+21i+96i-72}{16+9}$$

$$=\frac{-44+117i}{25}=-\frac{44}{25}+\frac{117}{25}i$$

So, $\operatorname{Re}(z)=-\dfrac{44}{25}$ and $\operatorname{Im}(z)=\dfrac{17}{25}$

Q3. **Prove that $\bar{z}=z$ iff z is real.**

Sol. Assume that $z=a+bi$

then $\bar{z}=\overline{a+bi}=a-bi$

Given $\bar{z}=z$

$a-bi=a+bi$

$a-bi-a-bi=0$

$-2bi=0 \Rightarrow b=0$

So, $z=a+0i=a$

Hence z is a real number if $\bar{z}=z$.

Conversely,

Assume that z is a real number.

$z=a \Rightarrow \bar{z}=\bar{a}=a=z$

Hence $\bar{z}=z$ if z is a real number.

Q4. **For $z\in C$, show that:**

(i) $\dfrac{z+\bar{z}}{2}=\operatorname{Re}(z)$

Sol. Take, L.H.S $=\dfrac{z+\bar{z}}{2}$

Assume that $z=a+bi$

So, L.H.S $=\dfrac{(a+bi)+(\overline{a+bi})}{2}$

$$=\frac{(a+bi)+(a-bi)}{2}=\frac{a+bi+a-bi}{2}=\frac{2a}{2}=a$$

L.H.S $=\operatorname{Re}(z)$

L.H.S = R.H.S

(ii) $\dfrac{z-\bar{z}}{2i} = \mathrm{Im}(z)$

Sol. $\text{L.H.S} = \dfrac{z-\bar{z}}{2i}$

As $z = a + bi$

So, $\text{L.H.S} = \dfrac{(a+bi)-\overline{(a+bi)}}{2i}$

$= \dfrac{(a+bi)-(a-bi)}{2i} = \dfrac{a+bi-a+bi}{2i} = \dfrac{2bi}{2i}$

$= b = \mathrm{Im}(z)$

L.H.S = R.H.S

(iii) $|z|^2 = z\cdot\bar{z}$

Sol. Assume that $z = a + bi$

$\text{L.H.S} = |z|^2$

$= |a+bi|^2 = \left(\sqrt{a^2+b^2}\right)^2$

$= a^2 + b^2$ ....(i)

$\text{R.H.S} = z.\bar{z}$

$= (a+bi)\cdot\overline{(a+bi)} = (a+bi)\cdot(a-bi)$

$= (a)^2 - (bi)^2 = a^2 - b^2i^2 = a^2 - b^2(-1)$

$= a^2 + b^2$ ....(ii)

From (i) and (ii)

L.H.S = R.H.S

Q5. **If $z_1 = 2+i$, $z_2 = 3-2i$, $z_3 = 1+3i$ then express $\dfrac{\bar{z}_1\bar{z}_3}{z_2}$ in the form of $a + ib$.**

Sol. As $z_1 = 2+i$, $z_2 = 3-2i$, $z_3 = 1+3i$

$\dfrac{\bar{z}_1\bar{z}_3}{z_2}$

$= \dfrac{\overline{(2+i)}\,\overline{(1+3i)}}{(3-2i)} = \dfrac{(2-i)(1-3i)}{3-2i}$

$= \dfrac{2(1-3i)-i(1-3i)}{3-2i} = \dfrac{2-6i-i+3i^2}{3-2i}$

$= \dfrac{2-6i-i+3(-1)}{3-2i} = \dfrac{2-6i-i-3}{3-2i} = \dfrac{-1-7i}{3-2i}$

$= \dfrac{-1-7i}{3-2i}\times\dfrac{3+2i}{3+2i} = \dfrac{-1(3+2i)-7i(3+2i)}{(3)^2-(2i)^2}$

$= \dfrac{-3-2i-21i-14i^2}{9-4i^2} = \dfrac{-3-2i-21i-14(-1)}{9-4(-1)}$

$= \dfrac{-3-2i-21i+14}{9+4} = \dfrac{11-23i}{13} = \dfrac{11}{13} - \dfrac{23}{13}i$

Q6. **If $z_1 = 2 + 7i$ and $z_2 = -5 + 3i$, then evaluate the following:**

(i) $|2z_1 - 4z_2|$

Sol. $|2z_1 - 4z_2| = |2(2+7i) - 4(-5+3i)|$

$= |4 + 14i + 20 - 12i| = |24 + 2i|$

$= \sqrt{24^2+2^2} = \sqrt{576+4} = \sqrt{580} = 2\sqrt{145}$

(ii) $|3z_1 + 2\bar{z}_1|$

Sol. $|3z_1 + 2\bar{z}_1| = \left|3(2+7i)+2\left(\overline{2+7i}\right)\right|$

$= |3(2+7i)+2(2-7i)|$

$= |6 + 21i + 4 - 14i| = |10 + 7i|$

$= \sqrt{10^2+7^2} = \sqrt{100+49} = \sqrt{149}$

(iii) $|-7z_2 + 2\bar{z}_2|$

Sol. $|-7z_2 + 2\bar{z}_2| = \left|-7(-5+3i)+2\left(\overline{-5+3i}\right)\right|$

$= |-7(-5+3i)+2(-5-3i)|$

$= |35 - 21i - 10 - 6i| = |25 - 27i|$

$= \sqrt{25^2+(-27)^2} = \sqrt{625+729} = \sqrt{1354}$

(iv) $|(z_1 + z_2)^3|$

Sol. $|(z_1 + z_2)^3|$

$= |(2 + 7i - 5 + 3i)^3| = |(-3 + 10i)^3|$

$= |(-3)^3 + 3(-3)^2(10i) + 3(-3)(10i)^2 + (10i)^3|$

$= |-27 + 3(9)(10i) + 3(-3)(100i^2) + 1000i^3|$

$= |-27 + 270i - 900i^2 + 1000i^3|$

$= |-27 + 270i - 900(-1) + 1000(-i)|$

$= |-27 + 270i + 900 - 1000i|$

$= |873 - 730i| = \sqrt{873^2+(-730)^2}$

$= \sqrt{762129+532900}$

$= \sqrt{1295029} = 109\sqrt{109}$

Q7. **Show that: $i^{n+1} + i^{n+2} + i^{n+3} + i^{n+4} = 0$, for all $n \in N$**

Sol. Proof:

$\text{L.H.S} = i^{n+1} + i^{n+2} + i^{n+3} + i^{n+4}$

$= i^n.i + i^n.i^2 + i^n.i^3 + i^n.i^4$

$= i^n(i + i^2 + i^3 + i^4)$

$= i^n(i + i^2 + i^2 + i + i^{2\times 2})$

$= i^n[i + (-1) + (-1).i + (-1)^2]$ $\quad\because i^2 = -1$

$= i^n(i - 1 - i + 1)$

$= i^n(0) = 0 = \text{R.H.S}$

$\because$ L.H.S = R.H.S

Q8. Find the least positive value of, n, if $\left(\frac{1+i}{1-i}\right)^{2n} = 1$.

Sol. $\left(\frac{1+i}{1-i}\right)^{2n} = 1$

$$\left(\frac{1+i}{1-i}\times\frac{1+i}{1+i}\right)^{2n} = 1$$

$$\left(\frac{1+i+i+i^2}{1^2-i^2}\right)^{2n} = 1$$

$$\left[\frac{1+i+i+(-1)}{1-(-1)}\right]^{2n} = 1$$

$$\left(\frac{1+i+i-1}{1+1}\right)^{2n} = 1$$

$$\left(\frac{2i}{2}\right)^{2n} = 1$$

$$i^{2n} = 1$$

$$i^{2n} = (-1)^2$$

$$i^{2n} = (i^2)^2$$

$$i^{2n} = i^4$$

By comparing both sides

$$2^n = 4$$

$$n = \frac{4}{2} \quad \Rightarrow \quad n = 2$$

Q9. Show that, the value of i^n for n $\in$ N and n > 4 is i^r, where r is the remainder when n is divided by 4.

Sol. Proof:

As, given n $\in N$ and n > 4. Also, r is the remainder when n is divided by 4.

So, we can write that $n = 4k + r$ where n $\in N$.

$$i^n = i^{4k+r}$$

$$i^n = (i)^{4k}.i^r$$

$$i^n = (1)^k.i^r \qquad \because i^4 = (1)$$

$$i^n = (1).i^r$$

$$i^n = i^r$$

Hence, proved.

1.2 Equality of Two Complex Numbers

Example 1: If $(3+2i)(x+iy) = 5 + 12i$, where $x,y \in R$, then find the values of x and y. Complex Numbers

Sol. Given that $(3 + 2i)(x + iy) = 5 + 12i$

$\Rightarrow$ $3x + 3iy + 2ix + 2yi^2 = 5 + 12i$

$\Rightarrow$ $(3x - 2y) + (2x + 3y)i = 5 + 12i$

Comparing real and imaginary part, we have

$3x - 2y = 5$ ...(i)

$2x + 3y = 12$ ...(ii)

Multiplying equation (i) by 3 and equation (ii) by 2, we have

$$9x - 6y = 15$$

$$4x + 6y = 24$$

Add the equations

$$9x - 6y + 4x + 6y = 15 + 24$$

$$13x = 39$$

$$x = 3$$

Substitute $x = 3$ in equation (i), we have

$$3(3) - 2y = 5$$

$$9 - 2y = -12$$

$$-2y = -4$$

$$y = 2$$

Thus, $x = 3, y = 2$

Example 2: Find the square root of complex number $5+12i$ and also represent the square roots on an Argand diagram.

Sol. Let $x + yi = 5 + 12i$

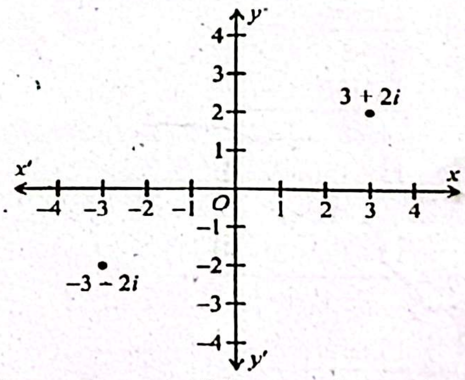
$\Rightarrow$ $x = 5$ and $y = 12 > 0$

$$|z| = |5+12i| = \sqrt{5^2+12^2} = 13,$$

Applying the square root formula for complex numbers, we get

$$\sqrt{5+12i} = \pm\left(\sqrt{\frac{13+5}{2}} + \frac{i12}{|12|}\sqrt{\frac{13-5}{2}}\right)$$

$$= \pm\left(\sqrt{9} + i\sqrt{4}\right) = \pm(3+2i)$$



Thus, the square root of the complexnumber $5 + 12i$ are $3 + 2i$ and $-3 - 2i$ are shown in adjacent figure.

Exercise 1.2

Q1. Find the values of x and y in each of the following:

(i) $x + iy + 2 - 3i = i(5 - i)(3 + 4i)$

Sol. $x + iy + 2 - 3i = i[5(3 + 4i) - i(3 + 4i)]$

$x + iy + 2 - 3i = i[15 + 20i - 3i - 4i^2]$